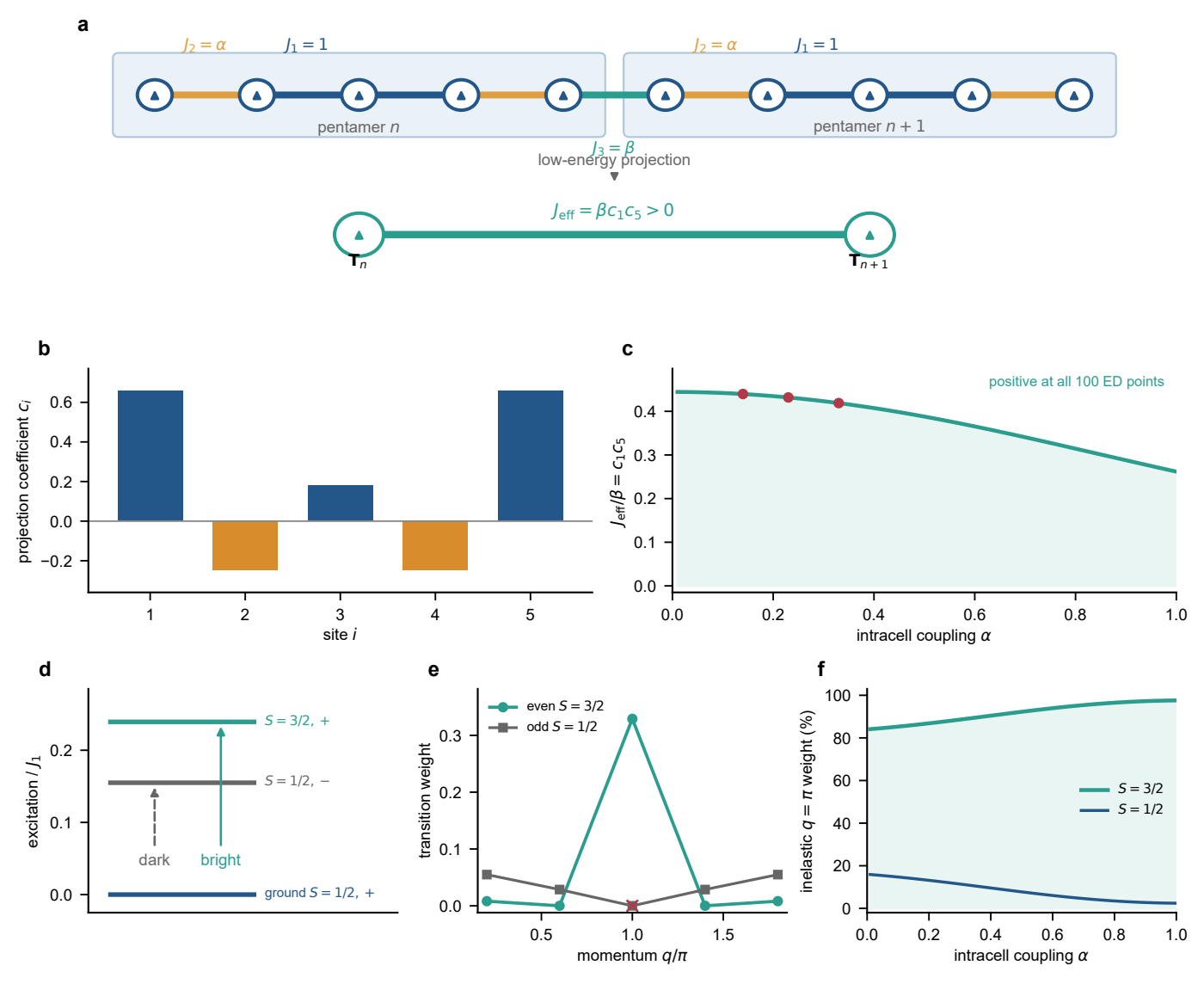

a
$J_2 = \alpha$
$J_1 = 1$
pentamer n
pentamer $n+1$
$J_3 = \beta$
low-energy projection
$J_{\text{eff}} = \beta c_1 c_5 > 0$
$\mathbf{T}_n$
$\mathbf{T}_{n+1}$
b
projection coefficient c_i
site i
c
positive at all 100 ED points
$J_{\text{eff}}/\beta = c_1 c_5$
intracell coupling α
d
excitation / J_1
$S = 3/2,\ +$
$S = 1/2,\ -$
dark
bright
ground $S = 1/2,\ +$
e
even $S = 3/2$
odd $S = 1/2$
transition weight
momentum q/π
f
inelastic $q = \pi$ weight (%)
$S = 3/2$
$S = 1/2$
intracell coupling α